

Shark Evolution Experiment

GA + RL run — 2026-06-09T11:19:36

Summary

Summary	
Generations	10000
Population (sharks/gen)	30
Best fitness	115.600
Final mean fitness	18.027
Analytical fitness (best genome)	-0.0756
Runtime	991.5 s

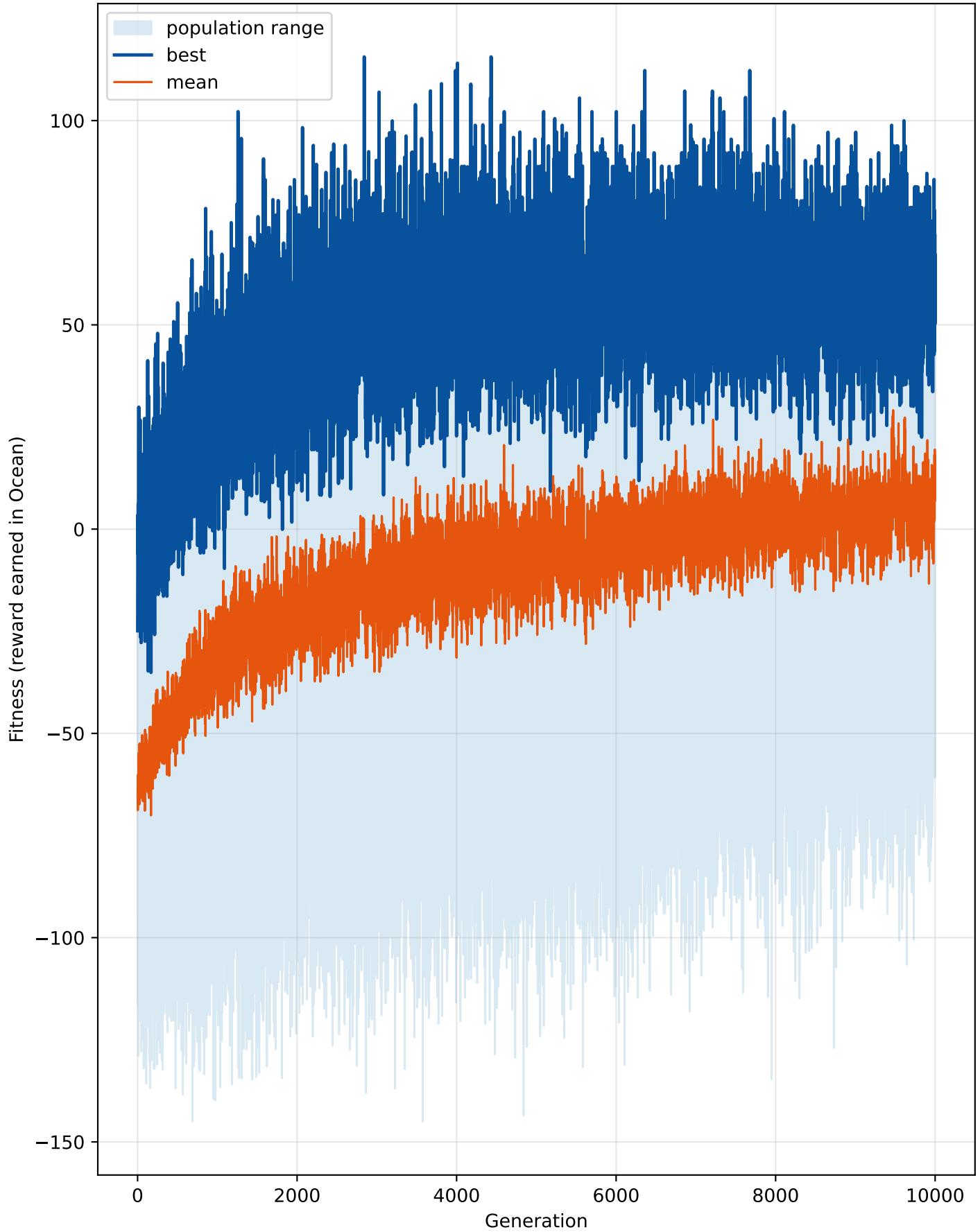
Run parameters

Run parameter	Value
Max steps per life	200
Lives averaged per genome	3
Mutation rate	0.2
Random seed	0
Run timestamp	2026-06-09T11:19:36

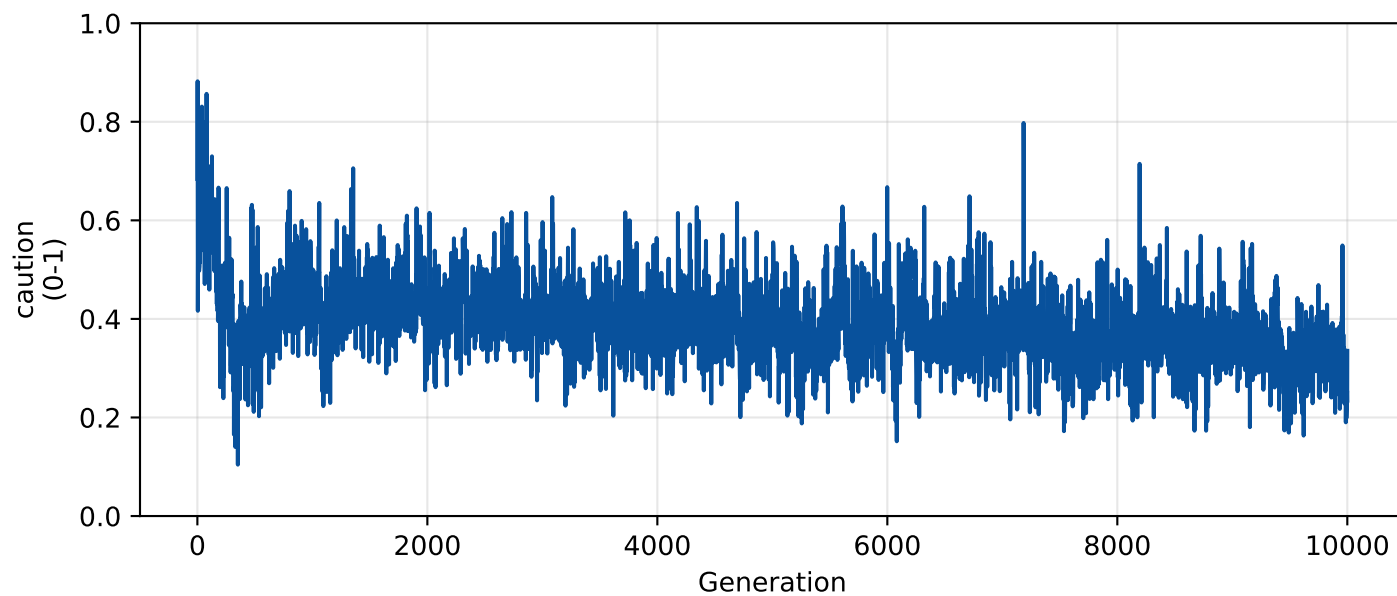
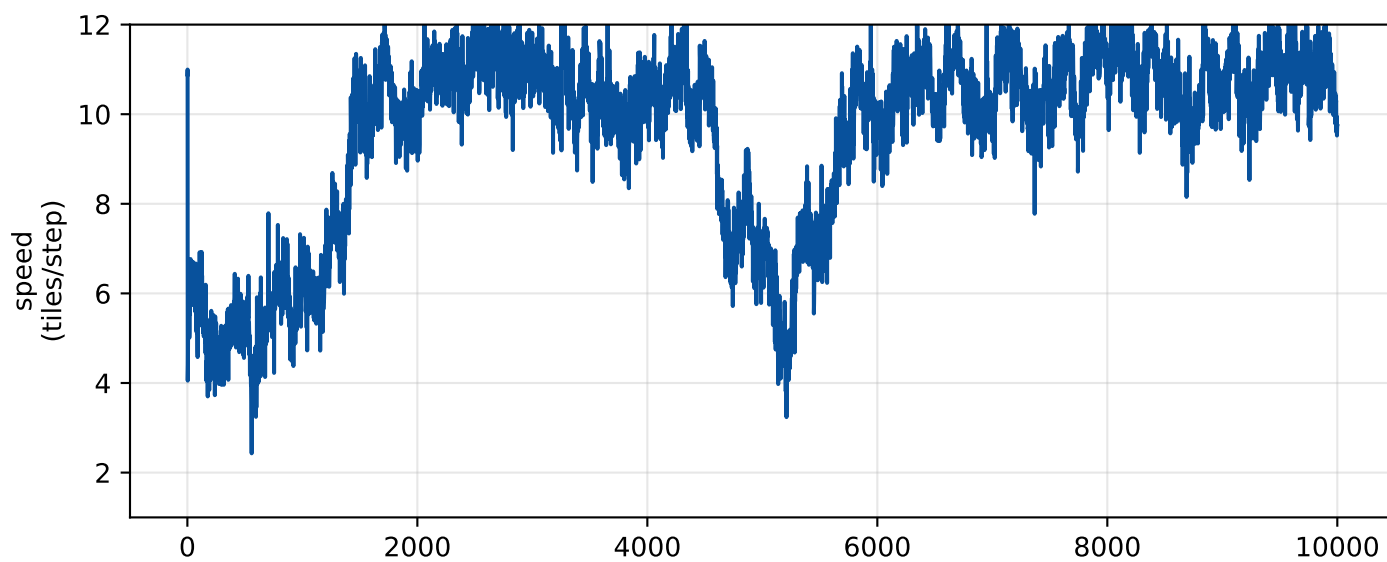
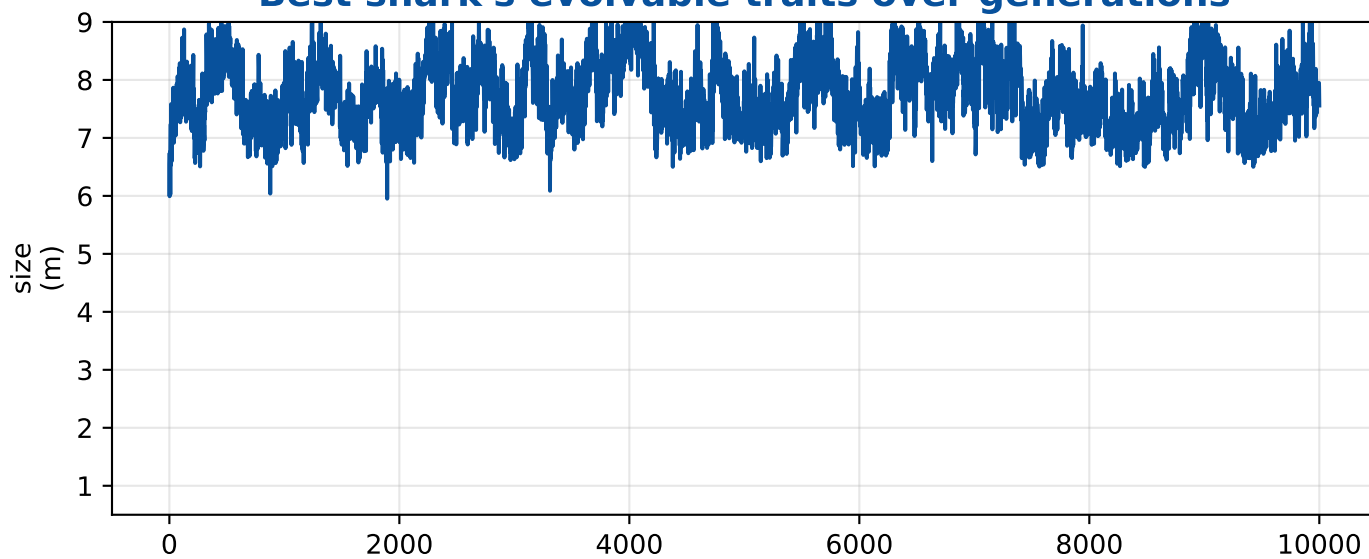
Shared brain (Q-learning)

Shared brain (Q-learning)	Value
States learned	237
Final epsilon (explore rate)	0.050
Epsilon min / decay	0.050 / 0.9997
Alpha (learning rate)	0.100
Gamma (discount)	0.950

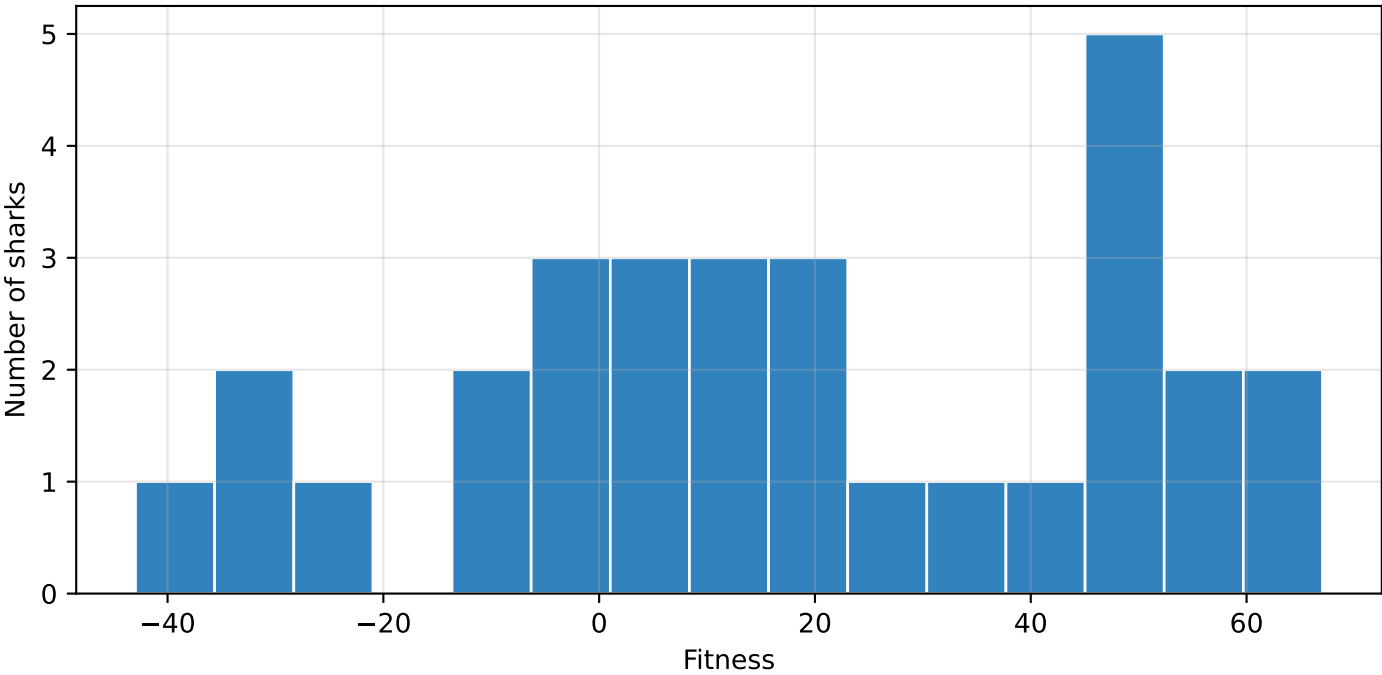
Fitness over generations



Best shark's evolvable traits over generations



Final-generation fitness distribution (gen 10000)



Best genome found (judged by reward earned in the Ocean)

Trait	Value	Unit	Evolved?
size	7.928	m	yes
speed	10.361	tiles/step	yes
caution	0.339	0-1	yes
gestation_period	12.000	months	no (fixed)
aggression	0.500	0-1	no (fixed)
field_of_perception	5.000	tiles (radius)	no (fixed)
color_hue	205.000	degrees (HSV hue)	no (fixed)

Per-generation summary (showing 31 of 10000 generations, every 334)

Gen	Size	Speed	Caution	Best	Mean	Min	Max
1	6.704	10.887	0.684	-24.900	-66.113	-110.900	-24.900
335	8.336	5.184	0.215	-1.133	-57.444	-101.033	-1.133
669	7.243	4.665	0.458	18.267	-36.573	-80.567	18.267
1003	7.858	6.348	0.454	23.100	-39.570	-113.700	23.100
1337	8.257	7.063	0.416	39.267	-20.301	-61.467	39.267
1671	7.174	9.886	0.410	38.733	-28.029	-88.433	38.733
2005	7.558	9.864	0.450	38.833	-17.362	-72.100	38.833
2339	8.292	11.951	0.395	54.133	-4.430	-50.433	54.133
2673	7.532	11.449	0.475	9.833	-32.301	-84.767	9.833
3007	7.241	10.396	0.435	21.233	-25.776	-103.733	21.233
3341	7.316	10.372	0.428	74.800	-17.358	-104.200	74.800
3675	9.000	10.951	0.333	41.900	-14.784	-73.300	41.900
4009	8.699	10.464	0.375	41.900	-23.532	-98.900	41.900
4343	7.627	11.159	0.627	58.633	0.340	-55.833	58.633
4677	7.337	6.951	0.471	39.900	-9.402	-76.000	39.900
5011	7.525	6.567	0.497	30.367	-25.912	-84.733	30.367
5345	7.314	7.123	0.360	73.700	-11.941	-71.700	73.700
5679	8.081	10.016	0.279	48.267	-6.436	-76.400	48.267
6013	7.062	9.478	0.333	50.267	-4.427	-91.300	50.267
6347	8.577	11.955	0.346	54.133	3.198	-76.600	54.133
6681	8.092	11.331	0.443	50.433	-6.299	-45.200	50.433
7015	7.734	10.213	0.363	48.700	-0.967	-61.567	48.700
7349	8.011	9.706	0.325	37.100	-12.401	-65.200	37.100
7683	7.991	10.478	0.423	35.267	-7.193	-90.833	35.267
8017	7.379	10.976	0.371	52.033	-4.908	-85.667	52.033
8351	7.323	10.921	0.383	60.333	9.276	-46.667	60.333
8685	6.942	9.883	0.353	42.800	3.426	-54.233	42.800
9019	8.211	10.677	0.349	50.233	-0.519	-60.800	50.233
9353	7.363	10.966	0.360	62.000	4.660	-78.200	62.000
9687	8.343	10.974	0.374	51.867	-3.023	-76.200	51.867
10000	7.559	9.740	0.335	67.033	18.027	-42.967	67.033